

# Brainstorming & Idea Prioritization

| Field         | Details                                                     |
|---------------|-------------------------------------------------------------|
| Team ID       | SWTID-2026-7253                                             |
| Team Size     | 5                                                           |
| Project       | OptiCrop: Smart Agricultural Production Optimization Engine |
| Team Leader   | Bommidi Deepika                                             |
| Team Member 1 | G Uday Kumar                                                |
| Team Member 2 | Kavali Anu                                                  |
| Team Member 3 | Pavan Kumar Sowdepalli                                      |
| Team Member 4 | Dasarayyagari Sravan Kumar                                  |

## 1. Brainstorming

The project team conducted a structured brainstorming session to identify real-world problems that can be effectively solved using Artificial Intelligence and Machine Learning in agriculture. Each proposed idea was evaluated based on practical applicability, technical feasibility, data availability, business value, implementation complexity, innovation potential, scalability, and long-term impact. After detailed discussion and comparison, the team selected the most suitable project for development.

## 3. Idea Evaluation Criteria

## 2. Brainstormed Ideas

| S.No | Project Idea              | Problem Solved                                                                   | Technical Feasibility | Business Impact | Priority |
|------|---------------------------|----------------------------------------------------------------------------------|-----------------------|-----------------|----------|
| 1    | OptiCrop                  | Recommends suitable crops using N, P, K, temperature, humidity, pH and rainfall. | High                  | High            | 1        |
| 2    | Smart Irrigation          | Optimizes irrigation using weather and soil moisture.                            | High                  | High            | 2        |
| 3    | Plant Disease Detection   | Detects crop diseases using AI.                                                  | High                  | Medium          | 3        |
| 4    | Fertilizer Recommendation | Suggests fertilizers based on soil nutrients.                                    | High                  | High            | 4        |
| 5    | Crop Yield Prediction     | Predicts crop production using historical data.                                  | Medium                | High            | 5        |

| Evaluation Criterion      | Importance | OptiCrop  |
|---------------------------|------------|-----------|
| Real-world Problem        | Very High  | Excellent |
| Data Availability         | High       | Excellent |
| ML Suitability            | Very High  | Excellent |
| Implementation Complexity | Medium     | Moderate  |
| Business Value            | Very High  | Excellent |
| Innovation                | High       | Very Good |
| Scalability               | High       | Excellent |
| Future Enhancement        | Very High  | Excellent |

## 4. Idea Prioritization

After evaluating all shortlisted ideas, OptiCrop: Smart Agricultural Production Optimization Engine was selected as the final project. It addresses real-world agricultural challenges by recommending suitable crops based on soil nutrients and environmental conditions. The project offers high business value, strong Machine Learning applicability, good dataset availability, excellent scalability, and opportunities for future enhancement.

## 5. Reasons for Selecting the Project

- Solves a real-world agricultural problem.
- Provides intelligent crop recommendations.
- Improves crop yield and resource efficiency.
- Supports sustainable farming.
- Uses AI and Machine Learning effectively.
- Has suitable agricultural datasets.
- Supports scalable web implementation.
- Can be enhanced with weather, fertilizer and disease prediction.

## Conclusion

The brainstorming and idea prioritization process ensured that OptiCrop aligns with academic objectives, technical feasibility, industrial relevance, and societal impact. It was selected because of its intelligent crop recommendation capabilities and its potential to improve agricultural productivity through data-driven decision making.